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(19) **United States**(12) **Patent Application Publication**
Huang et al.(10) **Pub. No.: US 2025/0285943 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **LEADFRAME**(71) Applicant: **Diodes Incorporated**, Plano, TX (US)(72) Inventors: **Li-Ju Huang**, Plano, TX (US); **Michael Yimin Zhang**, Dallas, TX (US)(73) Assignee: **Diodes Incorporated**, Plano, TX (US)(21) Appl. No.: **19/215,181**(22) Filed: **May 21, 2025****Related U.S. Application Data**

(60) Division of application No. 17/803,180, filed on Aug. 31, 2022, which is a continuation-in-part of application No. 29/813,450, filed on Oct. 28, 2021, now Pat. No. Des. 969,763, which is a division of application No. 29/692,906, filed on May 29, 2019, now Pat. No. Des. 939,458, said application No. 17/803,180 is a continuation-in-part of application No. 29/813,451, filed on Oct. 28, 2021, now Pat. No. Des. 969,764, which is a division of application No. 29/692,906, filed on May 29, 2019, now Pat. No. Des. 939,458, said application No. 17/803,180 is a continuation-in-part of application No. 29/813,452, filed on Oct. 28, 2021, now Pat. No. Des. 985,518, which is a division of application No. 29/692,906, filed on May 29, 2019, now Pat. No. Des. 939,458, said application No. 17/803,180 is a continuation-in-part of application No. 29/814,007, filed on Nov. 2, 2021, now Pat. No. Des. 980,811, which is a division of application No. 29/699,936, filed on Jul. 30, 2019, now Pat. No. Des. 940,090, which is a continuation-in-part of application No. 29/692,906, filed on May 29, 2019, now Pat. No. Des. 939,458, said application No. 17/803,180 is a continuation-in-part of application No. 29/814,008, filed on Nov. 2, 2021, now abandoned, which is a division of application No. 29/699,936, filed on Jul. 30, 2019, now Pat. No. Des. 940,090, which is a continuation-in-part of application No. 29/692,906,

filed on May 29, 2019, now Pat. No. Des. 939,458, said application No. 17/803,180 is a continuation-in-part of application No. 29/814,009, filed on Nov. 2, 2021, now Pat. No. Des. 969,093, which is a division of application No. 29/699,936, filed on Jul. 30, 2019, now Pat. No. Des. 940,090, which is a continuation-in-part of application No. 29/692,906, filed on May 29, 2019, now Pat. No. Des. 939,458.

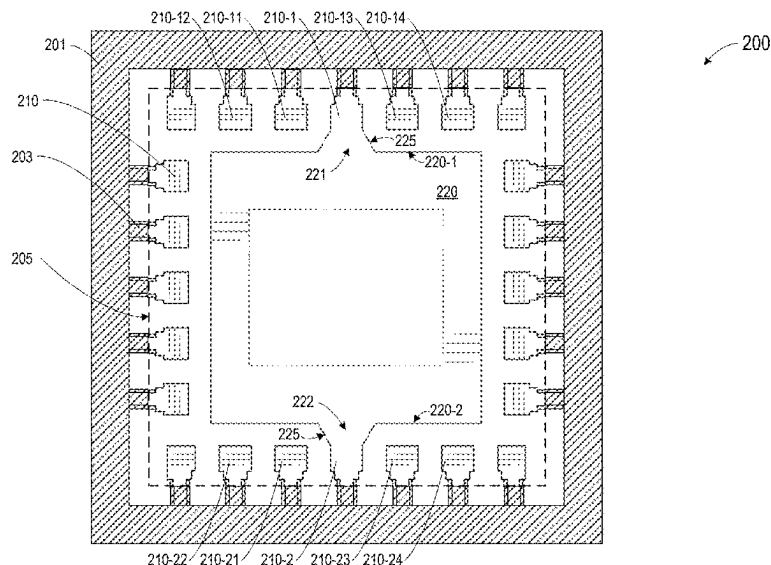
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(57)

ABSTRACT

A method for forming a semiconductor package includes providing a leadframe including a peripheral frame, a plurality of lead pads, and a die attach pad configured for mounting a semiconductor integrated circuit (IC) chip. The die attach pad comprises a first protruding portion on a first side of the die attach pad and a second protruding portion on a second side of the die attach pad. The method also includes attaching a first semiconductor IC die to the die attach pad, bonding IC die bonding pads to lead pads on the first side of the die attach pad, bonding IC die bonding pads to lead pads on the second side of the die attach pad, applying a molding that covers portions of the leadframe, a die attach material, and the first semiconductor IC die, and separating the plurality of lead pads from the peripheral frame.



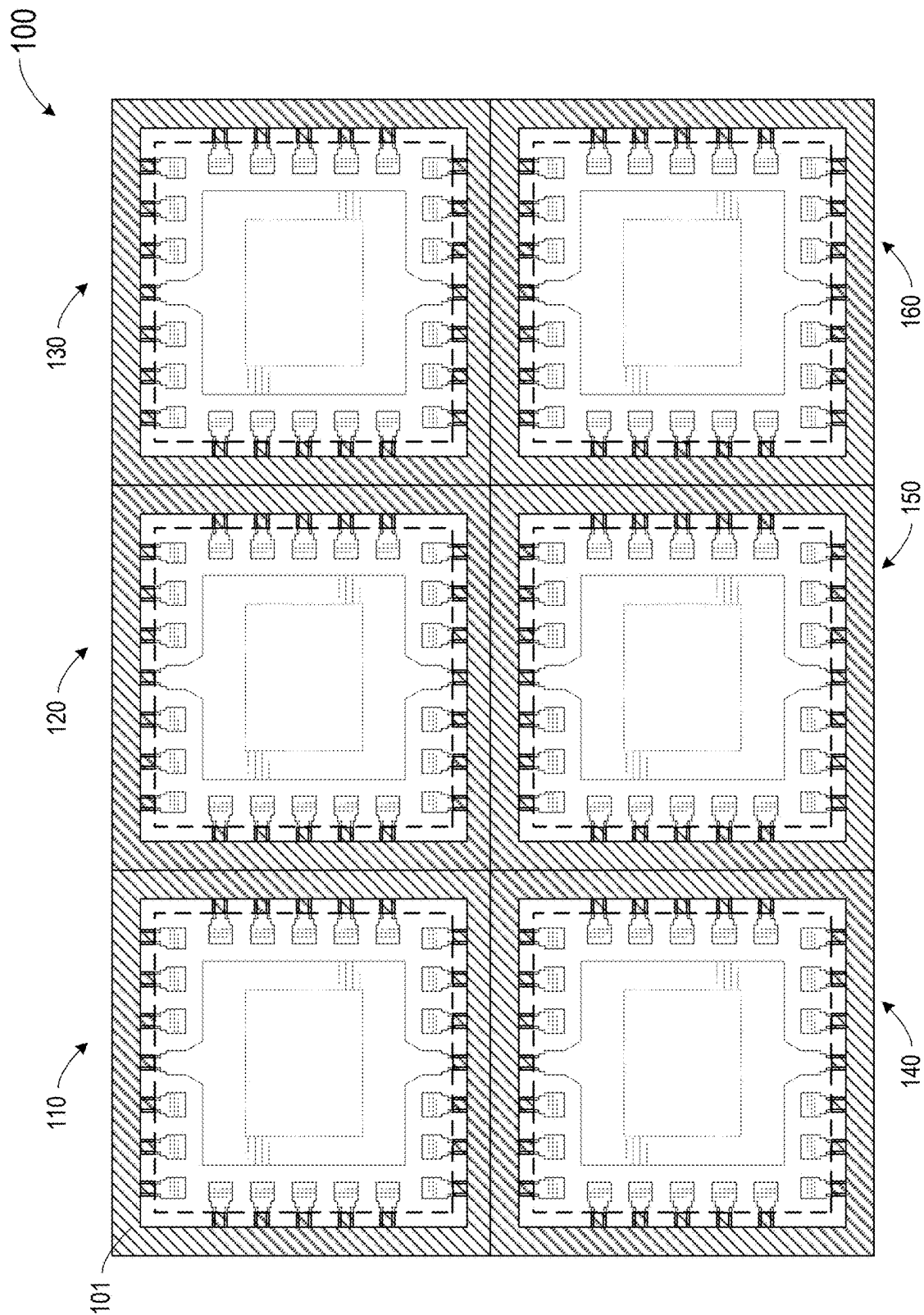
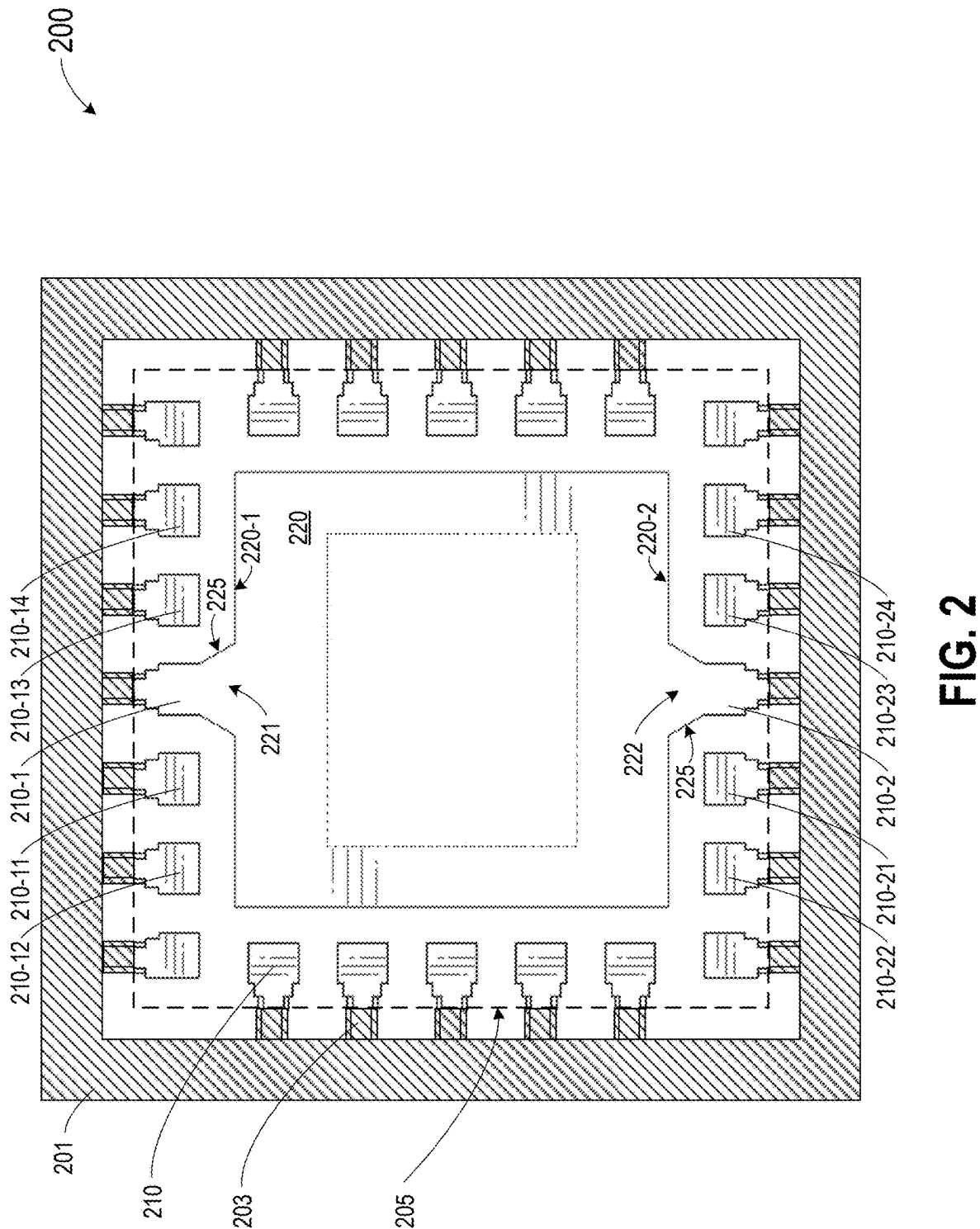


FIG. 1



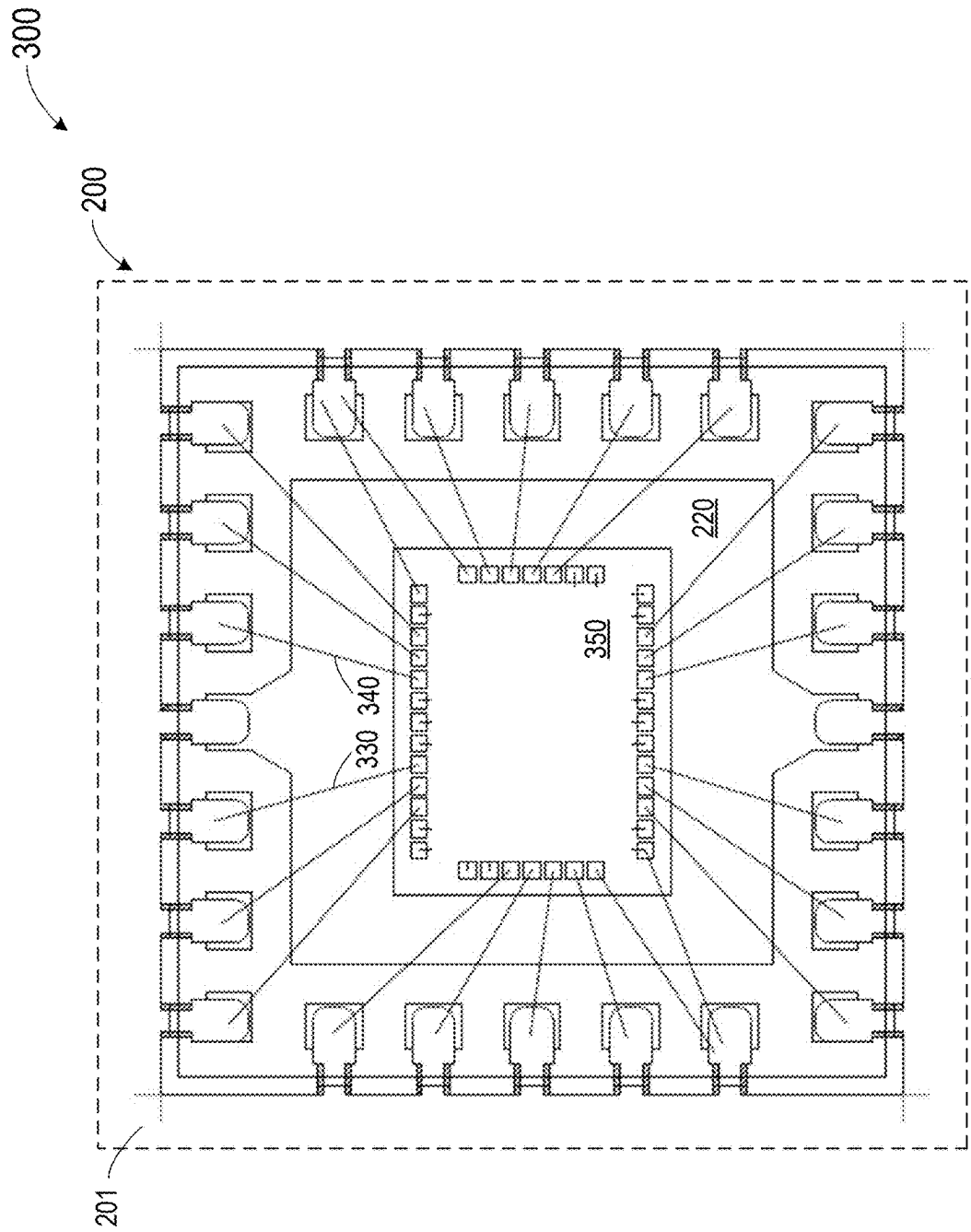


FIG. 3

400

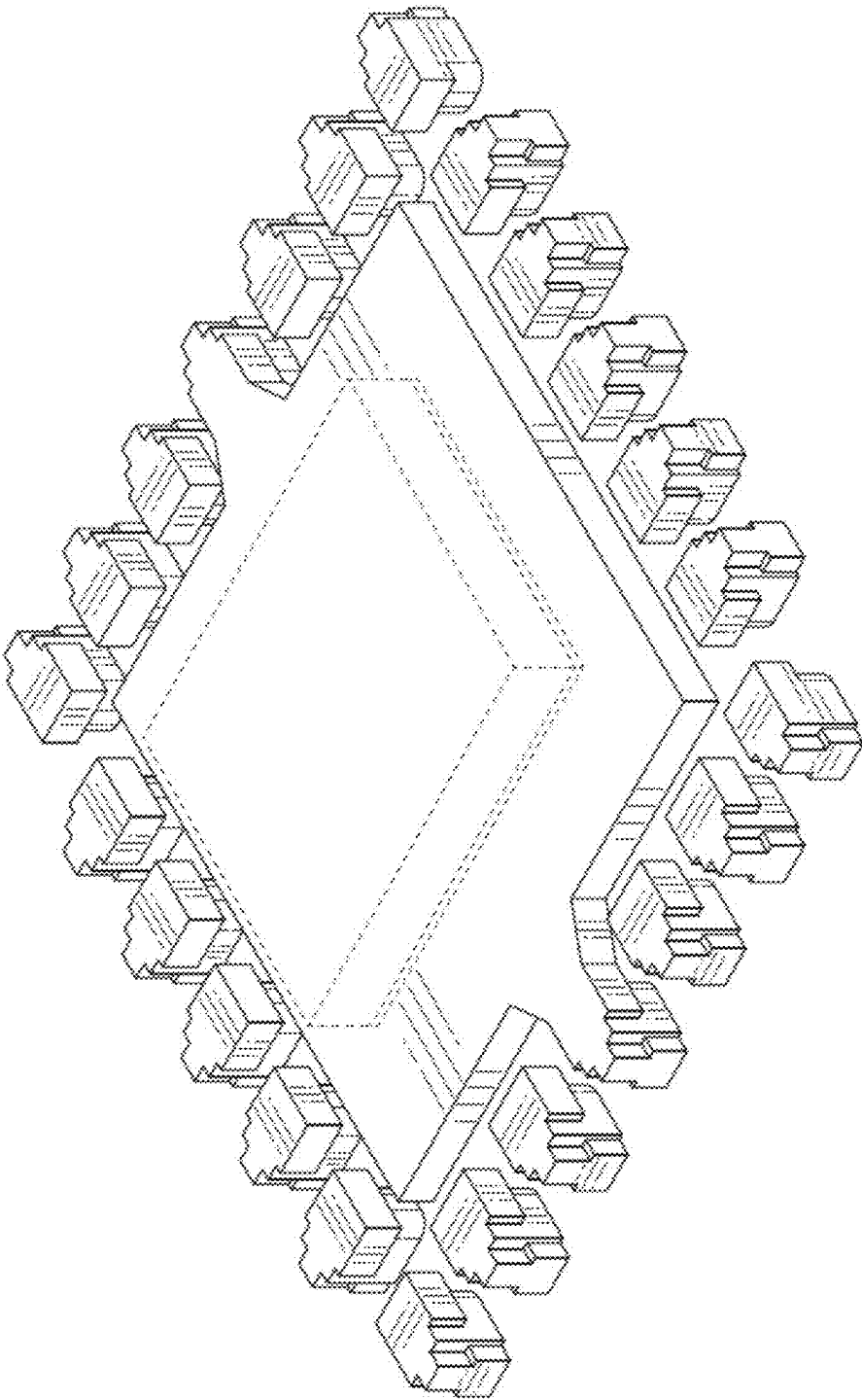


FIG. 4

500

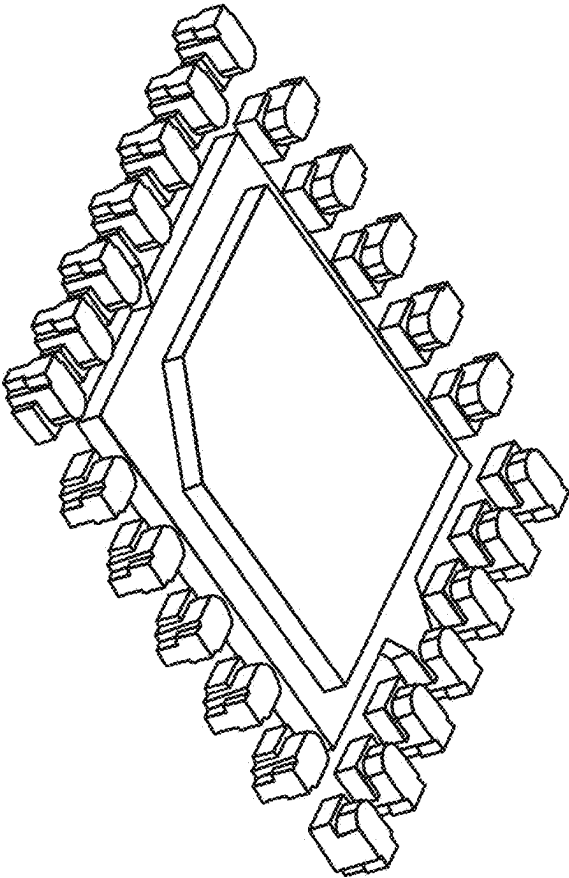


FIG. 5

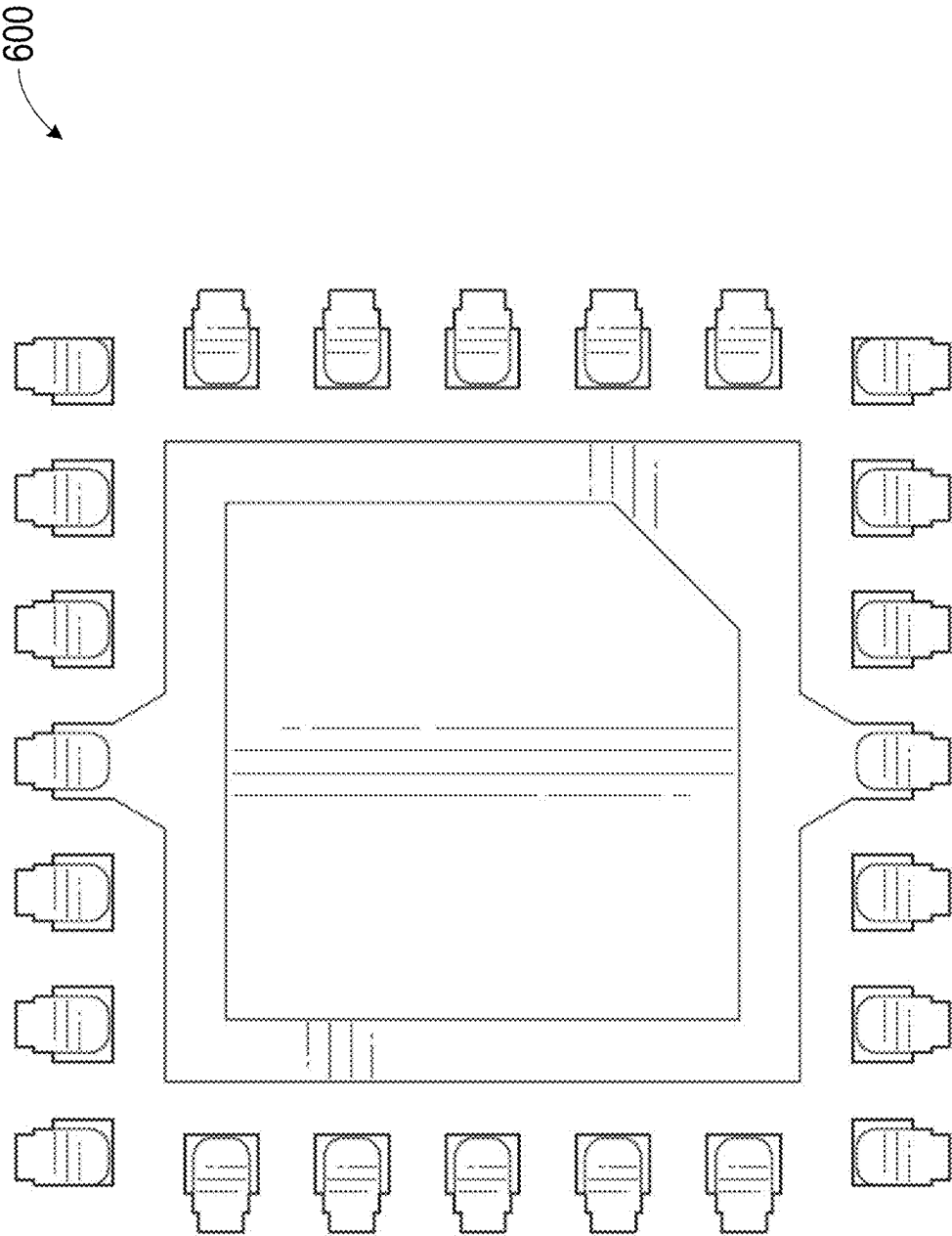


FIG. 6

700

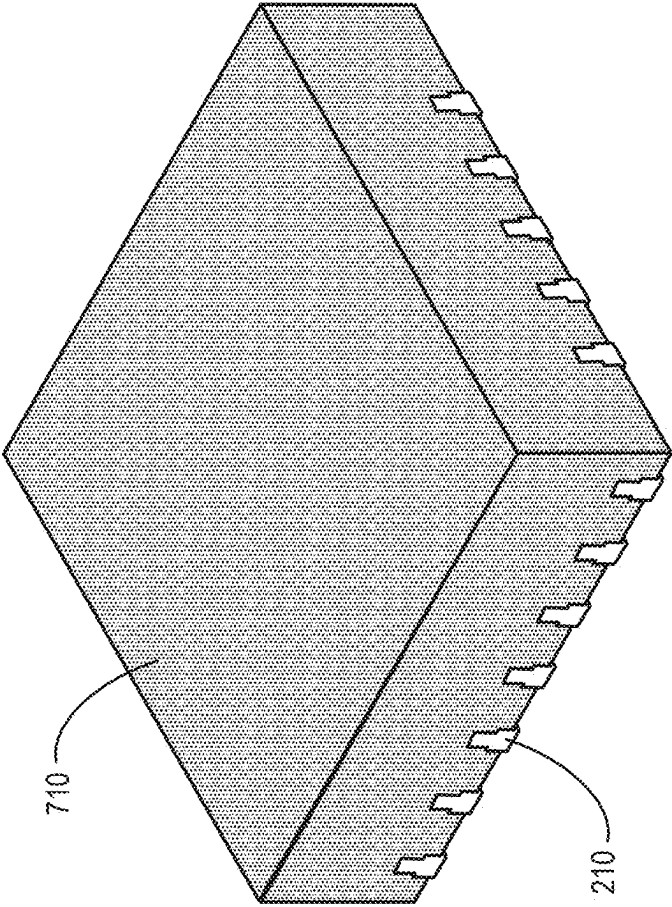


FIG. 7

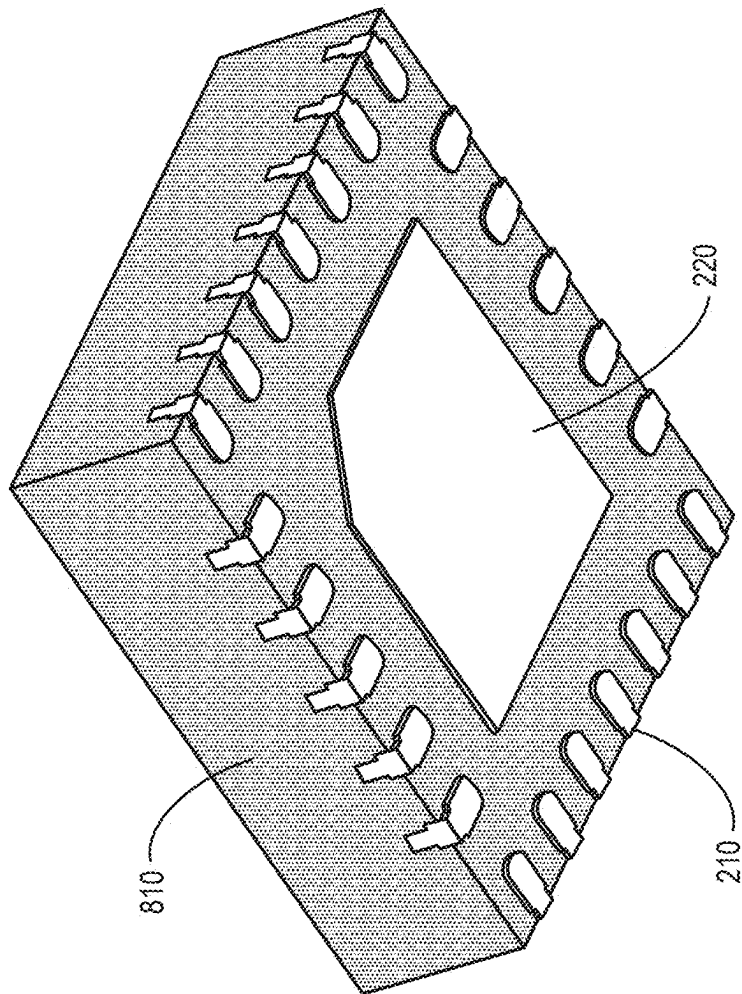


FIG. 8

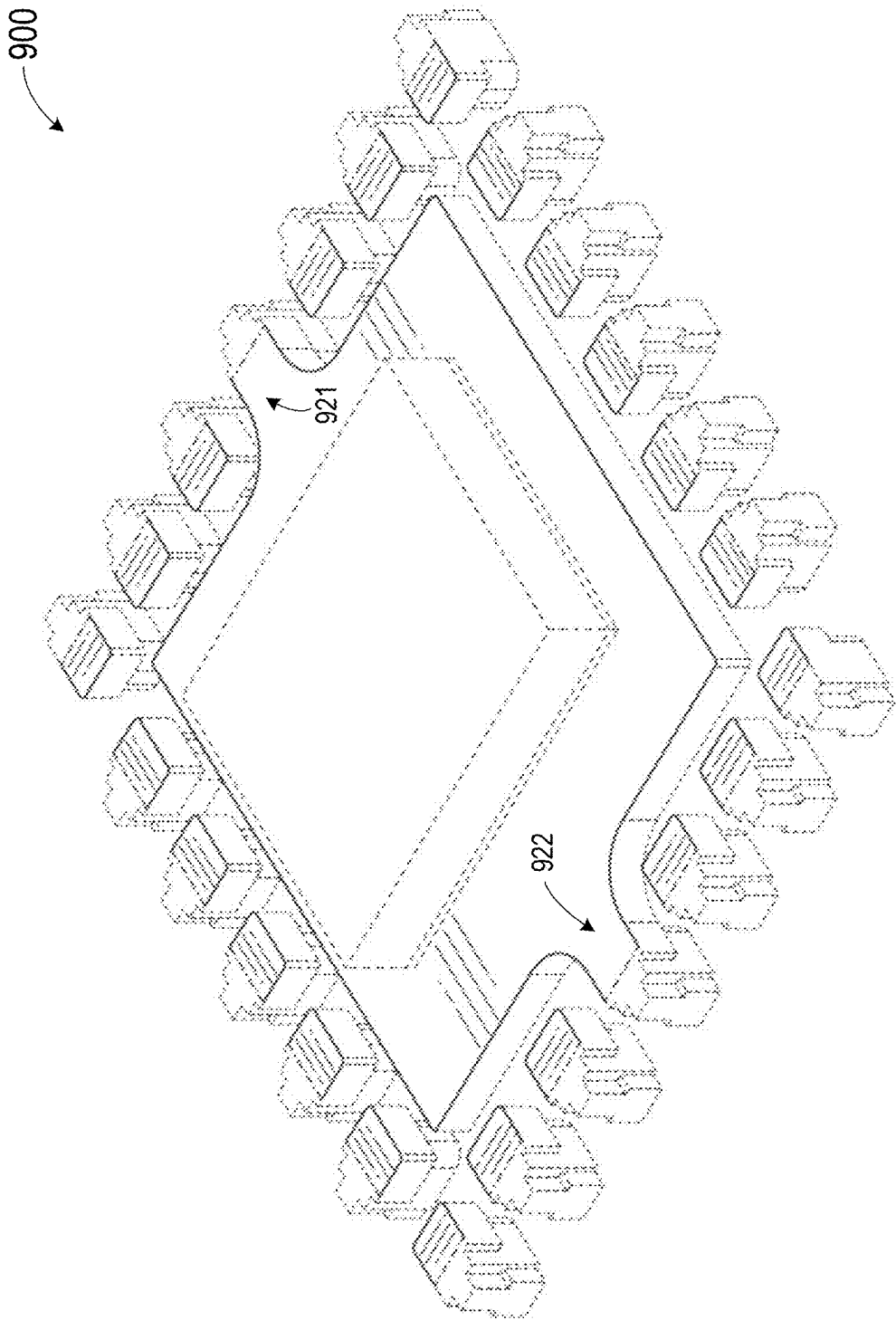


FIG. 9

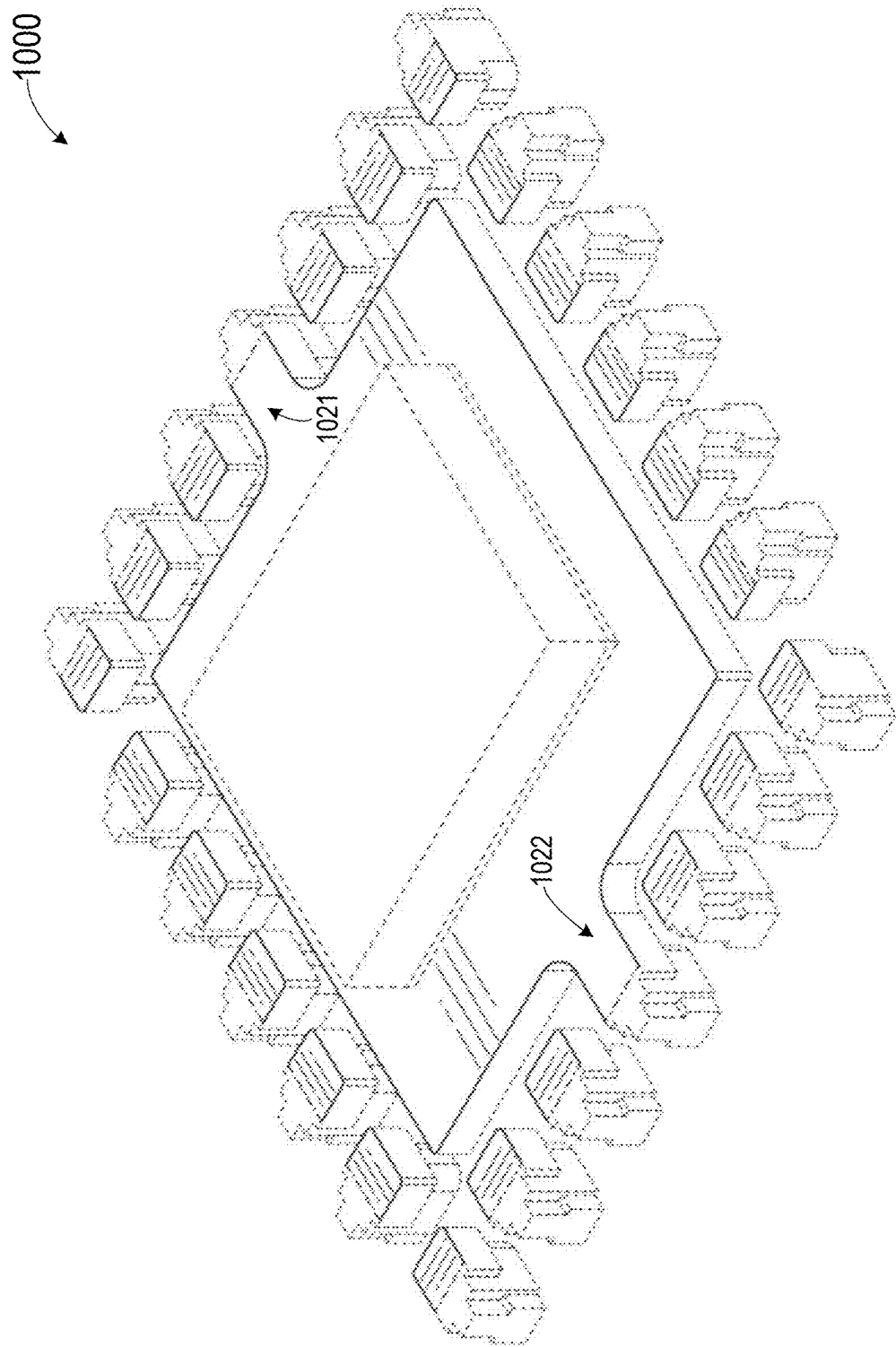


FIG. 10

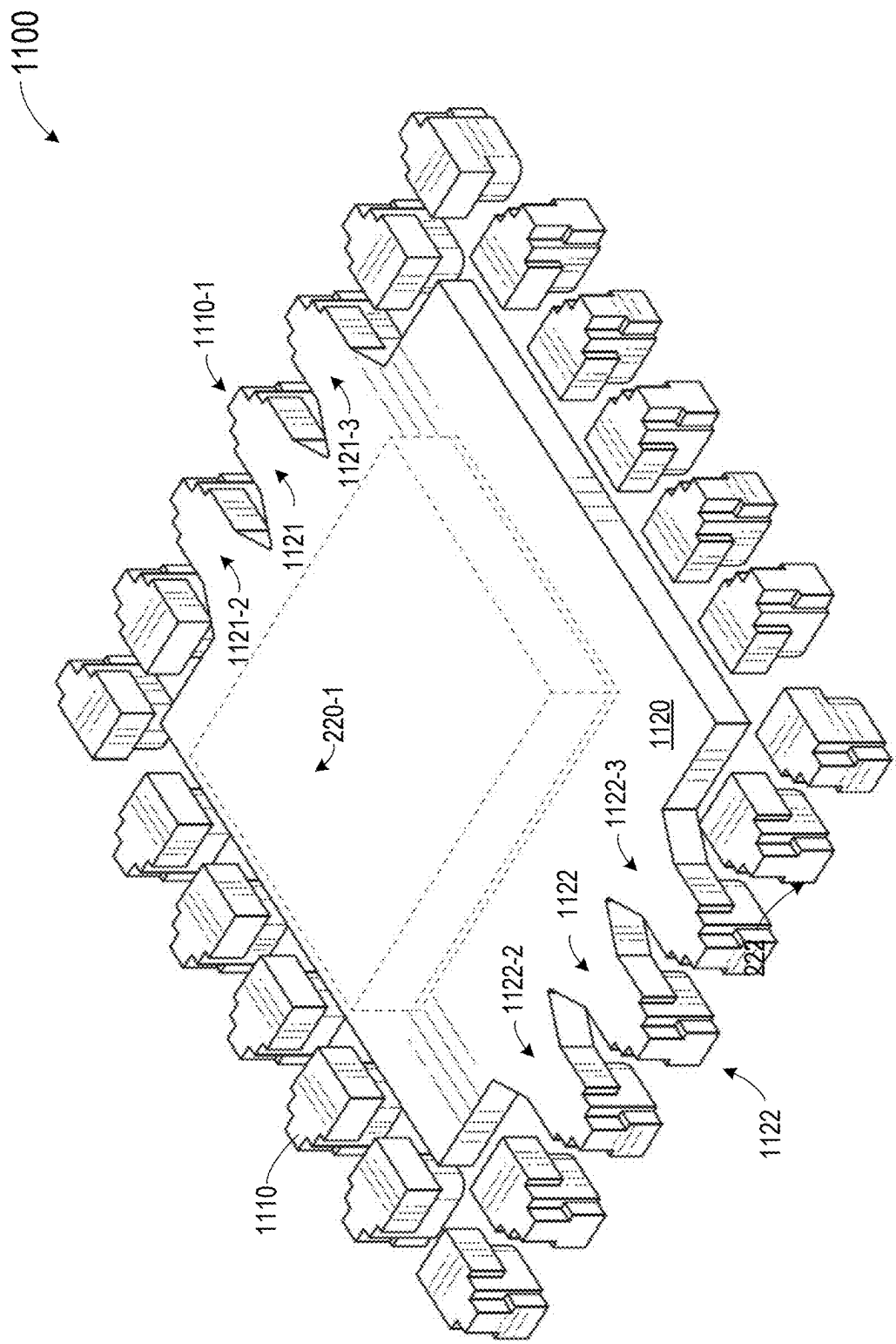


FIG. 11

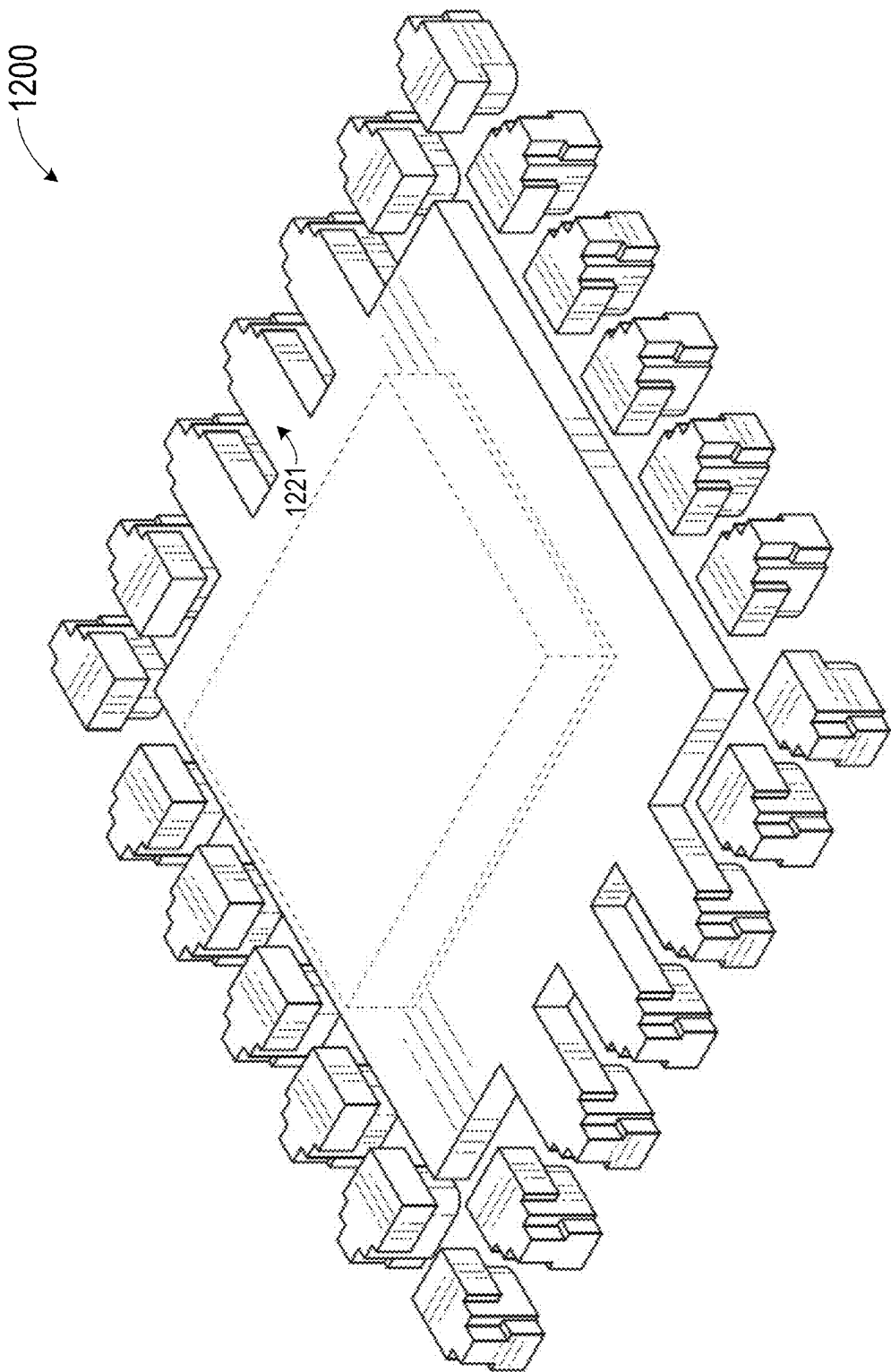


FIG. 12

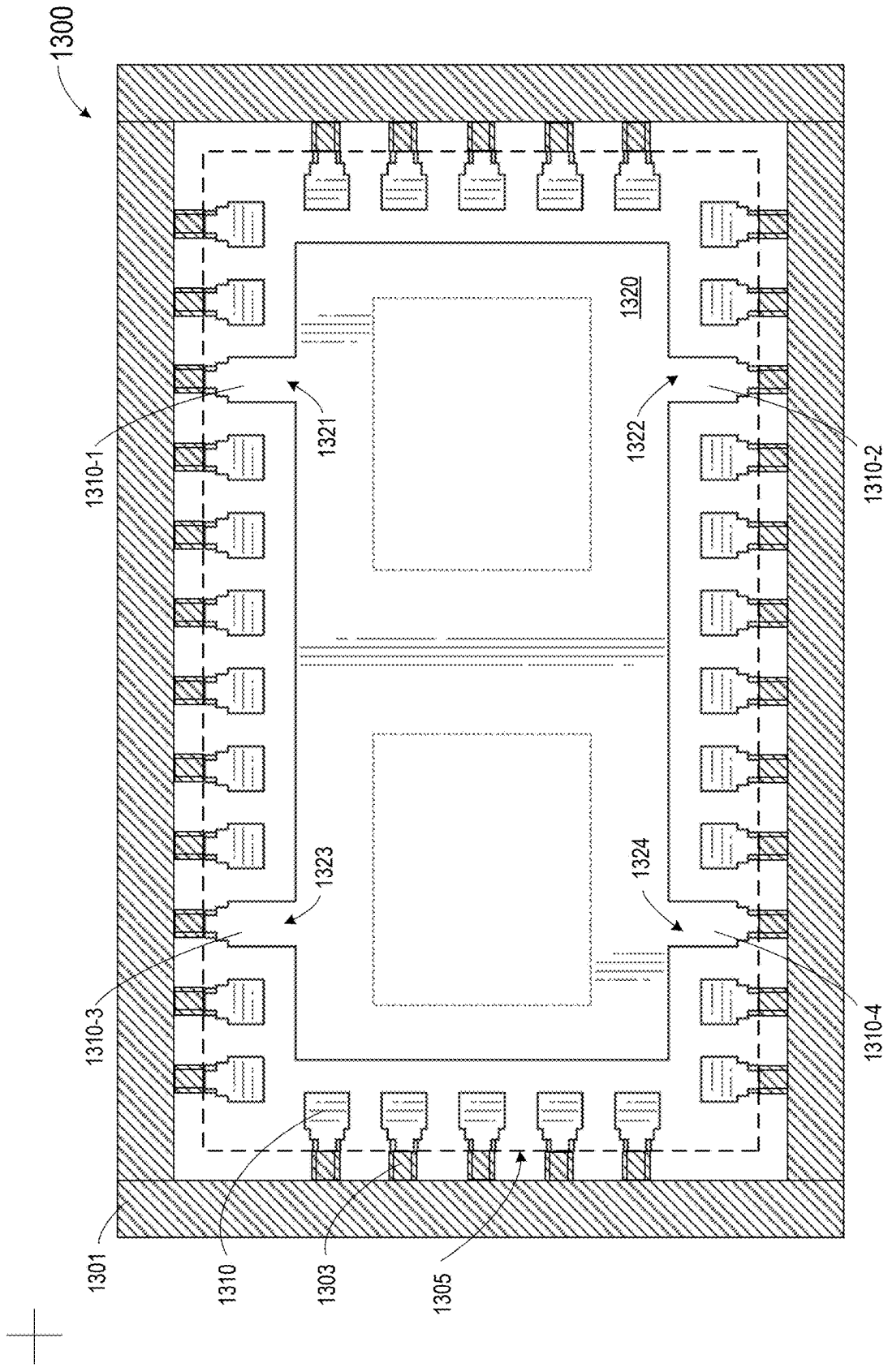


FIG. 13

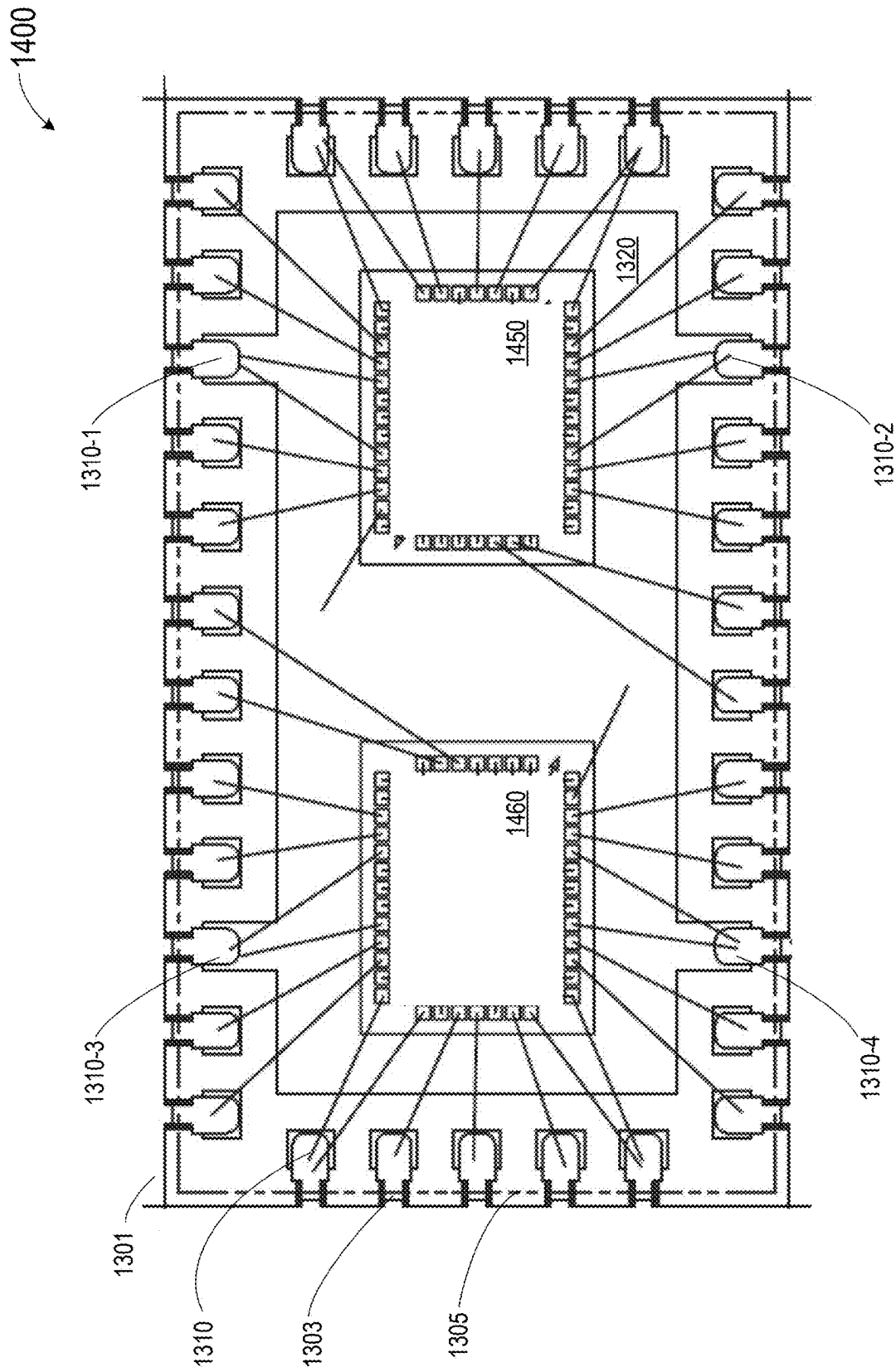


FIG. 14

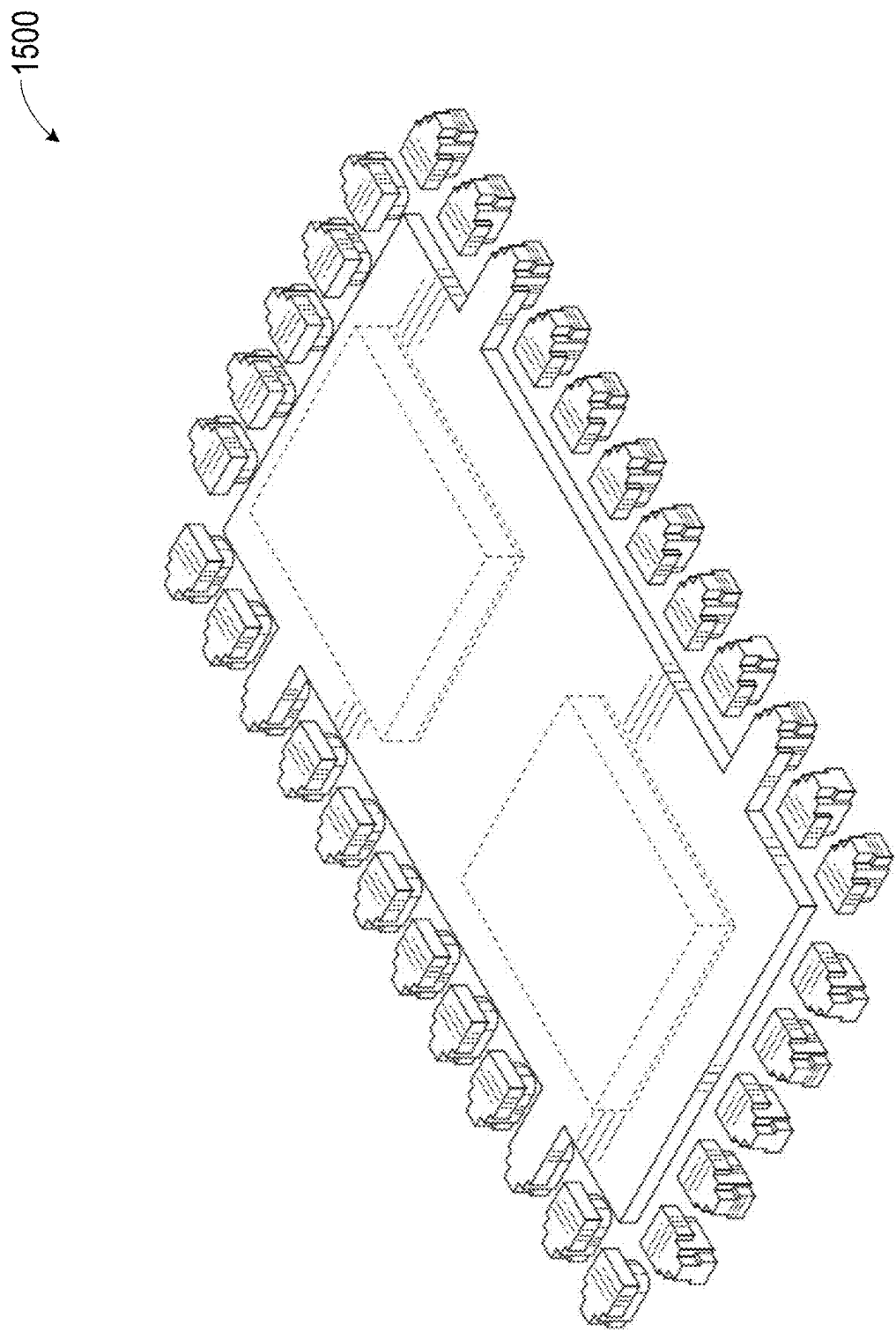


FIG. 15

1600

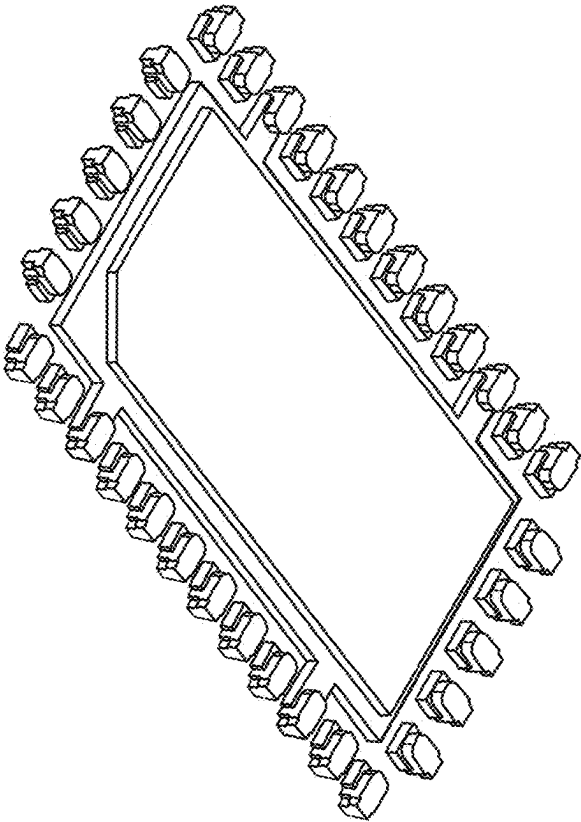


FIG. 16

1700

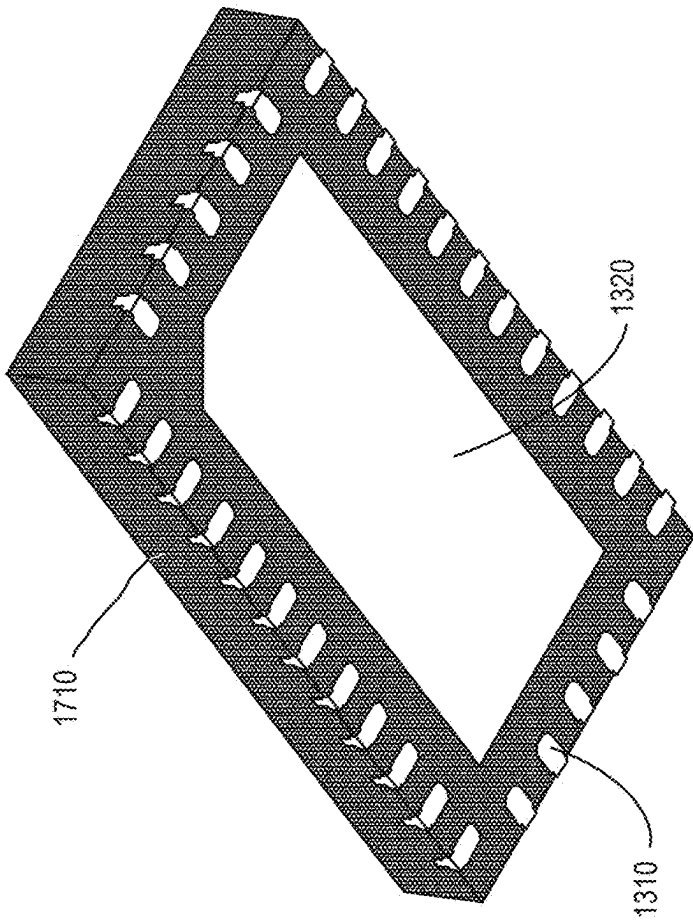


FIG. 17

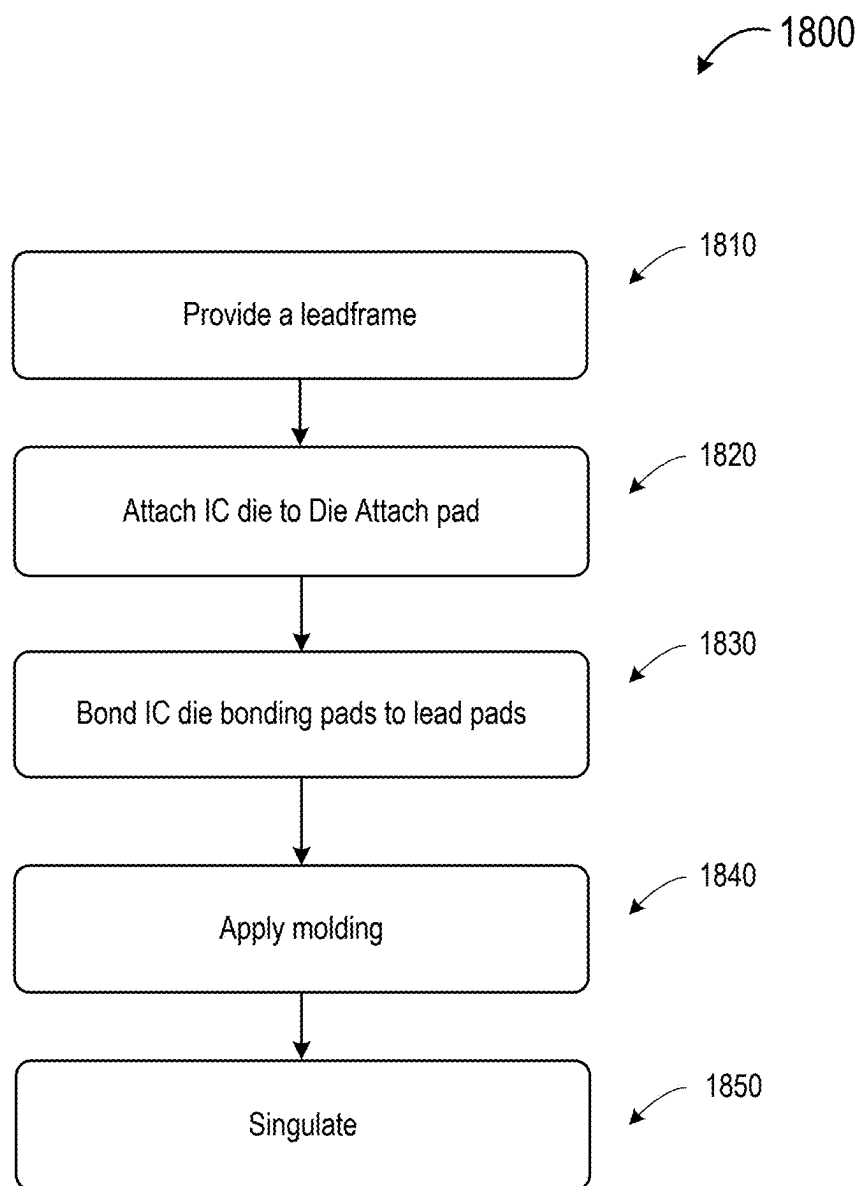


FIG. 18

LEADFRAME

CROSS-REFERENCES TO RELATED APPLICATIONS

[0001] This application is a division of U.S. patent application Ser. No. 17/803,180, entitled “LEADFRAME,” filed on Aug. 31, 2022, which is a Continuation-in-part application of U.S. Design Patent Application No. 29/813,450, entitled “LEADFRAME,” filed on Oct. 28, 2021, now U.S. Design Pat. No. D969763, U.S. Design Patent Application No. 29/813,451, entitled “LEADFRAME,” filed on Oct. 28, 2021, now U.S. Design Pat. No. D969764, and U.S. Design Patent Application No. 29/813,452, entitled “LEADFRAME,” filed on Oct. 28, 2021, now U.S. Design Pat. No. D985518, which are divisions of U.S. Design Patent Application No. 29/692,906, filed May 29, 2019, now U.S. Design Pat. No. D939,458. U.S. patent application Ser. No. 17/803,180 is also a Continuation-in-part application of U.S. Design Patent Application No. 29/814,007, entitled “LEADFRAME,” filed on Nov. 2, 2021, now U.S. Design Pat. No. D980811, U.S. Design Patent Application No. 29/814,008, entitled “LEADFRAME,” filed on Nov. 2, 2021, now abandoned, and U.S. Design Patent Application No. 29/814,009, entitled “LEADFRAME,” filed on Nov. 2, 2021, now U.S. Design Pat. No. D969093, which are divisions of U.S. Design Patent Application No. 29/699,936, filed Jul. 30, 2019, now U.S. Design Pat. No. D940,090, which is a Continuation-in-part of U.S. Design Patent Application No. 29/692,906, filed May 29, 2019, now U.S. Design Pat. No. D939,458. The disclosures of all of the foregoing applications are hereby incorporated by reference in their entirety for all purposes.

BACKGROUND

[0002] This invention relates to the field of semiconductor device packaging. More particularly, the invention relates to leadframe design techniques for improving device performances.

[0003] A semiconductor package encloses an integrated circuit (IC) to form packaged integrated circuits. An integrated circuit die is typically attached to a die attach pad, a metal leadframe having leads physically isolated from the die attach pad, bond wires which electrically connect pads on the integrated circuit die to individual leads of the leadframe, and a hard encapsulant material which covers part or all of the package components, forming the exterior of the package, protects the integrated circuit from the environments, and leaves leads exposed, for electrical connection between the die and a printed circuit board.

[0004] In general, IC packages are either leaded or leadless. A leaded package has metal lead conductors that extend from the package for attaching to a printed circuit board. In contrast, a leadless package has exposed metal leads on one surface of the outside of the package. The exposed leads are substantially coplanar with the outside surface of the package and the leads are soldered to a printed circuit board. The leadless package has a lower profile than the leaded package and consumes less space on the support board. Thus, leadless packages are often used when space is a premium as in small systems such as cellular telephones, personal digital assistants, and laptop computers.

[0005] Even though leadless IC packages are in wide use, there is still a need for improved packaging technologies.

DEFINITIONS

[0006] The terms used in this disclosure generally have their ordinary meanings in the art within the context of the invention. Certain terms are discussed below to provide additional guidance to the practitioners regarding the description of the invention. It will be appreciated that the same thing may be said in more than one way. Consequently, alternative language and synonyms may be used.

[0007] A semiconductor device chip or die in this paper refers to a slab of semiconductor material that contains an electronic component. In a discrete device, the chip may only contain a single circuit element such as a power transistor or a diode; in modern IC, a chip may contain over 10 billion circuit elements. Semiconductor material includes elements in Group IV on the periodic table, such as germanium, silicon, diamond, and compounds such as gallium arsenide, gallium nitride, silicon carbide, etc. Semiconductor chips are sawed from a finished semiconductor wafer; as such, they each have two opposite surfaces of major crystallographic planes, which are referred to in this paper as the top chip surface and the bottom surface.

[0008] When the phrase “the same” is used to describe two quantities, it means that the values of two quantities are determined the same within measurement or manufacturing limitations.

SUMMARY OF THE INVENTION

[0009] According to some embodiments, a leadframe includes a peripheral frame, a plurality of lead pads, and a die attach pad (DAP). Each lead pad is physically connected to the peripheral frame by a respective connecting portion. The DAP is surrounded by the plurality of lead pads. The DAP includes a first protruding portion coupled to a first lead pad on a first side of the DAP and a second protruding portion coupled to a second lead pad on a second side of the DAP opposite the first side. The DAP does not comprise direct connections to the peripheral frame. The leadframe further includes two or more of the lead pads disposed on either side of the first lead pad on the first side of the DAP; and two or more of the lead pads disposed on either side of the second lead pad on the second side of the DAP. In these embodiments, the first and second lead pads provide ground isolation and shielding for active signals to prevent interference and cross talk. Further, the die attach pad has no direct connection to the peripheral frame, simplifying the design and manufacturing of the leadframe.

[0010] According to some embodiments, a semiconductor package includes a leadframe. The leadframe has a die attach pad surrounded by a plurality of lead pads. The die attach pad is characterized by a rectangular shape and includes a first protruding portion on a first side of the die attach pad, the first protruding portion coupled to a first lead pad, and multiple lead pads on the first side of the die attach pad aligned with the first lead pad, with two or more of the lead pads disposed on both sides of the first lead pad. The die attach pad also includes a second protruding portion on a second side of the die attach pad opposite the first side, the second protruding portion coupled to a second lead pad; and multiple lead pads on the second side of the die attach pad aligned with the second lead pad, with two or more of the lead pads disposed on both sides of the second lead pad. The semiconductor package also includes a semiconductor integrated circuit (IC) die attached to the die attach pad of the

leadframe, wherein the die attach pad provides electrical ground for the first lead pad and the second lead pad. The semiconductor package also includes bonding wires coupling bonding pads for active signals on the IC die to lead pads on the first side of the die attach pad that are not coupled to the die attach pad; and bonding wires coupling bonding pads for active signals on the IC die to lead pads on the second side of the die attach pad that are not coupled to the die attach pad.

[0011] In some embodiments of the above semiconductor package, the leadframe further comprises a peripheral frame connected to the plurality of lead pads, wherein the die attach pad does not comprise a direct connection to the peripheral frame.

[0012] According to some embodiments, a leadframe includes a peripheral frame, a plurality of lead pads, and a die attach pad. Each of the plurality of lead pads is physically connected to the peripheral frame by a respective connecting portion. The die attach pad is configured for mounting a semiconductor integrated circuit (IC) chip, the die attach pad being surrounded by the plurality of lead pads. The die attach pad is characterized by a rectangular shape and includes a first protruding portion on a first side of the die attach pad, and the first protruding portion coupled to a first lead pad. The die attach pad also includes a second protruding portion on a second side of the die attach pad opposite the first side, the second protruding portion coupled to a second lead pad. The die attach pad does not comprise direct connections to the peripheral frame. The leadframe further includes multiple lead pads on the first side of the die attach pad, with two or more of the lead pads disposed on either side of the first lead pad; and multiple lead pads on the second side of the die attach pad, with two or more of the lead pads disposed on either side of the second lead pad.

[0013] According to some embodiments, a method for forming a semiconductor package includes providing a leadframe. The leadframe includes a peripheral frame, a plurality of lead pads, each of the plurality of lead pads physically connected to the peripheral frame by a respective connecting portion, and a die attach pad configured for mounting a semiconductor integrated circuit (IC) chip. The die attach pad is surrounded by the plurality of lead pads. The die attach pad is characterized by a rectangular shape and includes a first protruding portion and a second protruding portion. The first protruding portion is located on a first side of the die attach pad, and the first protruding portion is coupled to a first lead pad. The second protruding portion is located on a second side of the die attach pad opposite the first side, and the second protruding portion is coupled to a second lead pad. The die attach pad does not comprise direct connections to the peripheral frame. The method for forming a semiconductor package also includes attaching a first semiconductor integrated circuit (IC) die to the die attach pad, bonding IC die bonding pads for active signals to lead pads on the first side of the die attach pad that are not coupled to the die attach pad, and bonding IC die bonding pads for active signals to lead pads on the second side of the die attach pad that are not coupled to the die attach pad. The method for forming a semiconductor package also includes applying a molding that covers portions of the leadframe, the die attach material, and the die, and separating the plurality of lead pads from the peripheral frame by cutting off the connecting portions between the lead pads and the peripheral frame.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is a simplified schematic diagram illustrating a top view of a leadframe array according to some embodiments of the invention;

[0015] FIG. 2 is a simplified schematic diagram illustrating a top view of a leadframe according to some embodiments of the invention;

[0016] FIG. 3 is a simplified schematic diagram illustrating a top view of a semiconductor chip on a leadframe according to some embodiments of the invention;

[0017] FIG. 4 illustrates a top perspective view of a leadframe after singulation according to some embodiments of the invention;

[0018] FIG. 5 illustrates a bottom perspective view of a leadframe after singulation according to some embodiments of the invention;

[0019] FIG. 6 illustrates a bottom plan view of a leadframe after singulation according to some embodiments of the invention;

[0020] FIG. 7 illustrates a top perspective view of a semiconductor package according to some embodiments of the invention;

[0021] FIG. 8 illustrates a bottom perspective view of a semiconductor package according to some embodiments of the invention;

[0022] FIG. 9 illustrates a top perspective view of another leadframe after singulation according to some embodiments of the invention;

[0023] FIG. 10 illustrates a top perspective view of another leadframe after singulation according to some embodiments of the invention;

[0024] FIG. 11 illustrates a top perspective view of another leadframe after singulation according to some embodiments of the invention;

[0025] FIG. 12 illustrates a top perspective view of another leadframe after singulation according to some embodiments of the invention;

[0026] FIG. 13 is a simplified schematic diagram illustrating a top view of a leadframe according to some embodiments of the invention;

[0027] FIG. 14 is a simplified schematic diagram illustrating a top view of two semiconductor chips on a leadframe according to some embodiments of the invention;

[0028] FIG. 15 illustrates a top perspective view of a leadframe after singulation according to some embodiments of the invention;

[0029] FIG. 16 illustrates a bottom perspective view of a leadframe after singulation according to some embodiments of the invention;

[0030] FIG. 17 illustrates a bottom perspective view of a semiconductor package according to some embodiments of the invention; and

[0031] FIG. 18 is a simplified flowchart illustrating a method for operating an output driver according to some embodiments of the invention.

DETAILED DESCRIPTION

[0032] FIG. 1 is a simplified schematic diagram illustrating a top view of a leadframe array according to some embodiments of the invention. As shown in FIG. 1, leadframe array 100 includes a plurality of leadframes 110, 120, 130, 140, 150, and 160, etc. Leadframe array 100 has a

peripheral frame **101**, shown as a shaded region, that holds all the leadframes together until they are singulated.

[0033] FIG. 2 is a simplified schematic diagram illustrating a top view of a leadframe according to some embodiments of the invention. As shown in FIG. 2, a leadframe **200** includes a peripheral frame and a plurality of lead pads **210**. Each of the plurality of lead pads **210** is physically connected to the peripheral frame **201** by a respective connecting portion **203**. Leadframe **200** also has a die attach pad **220** configured for mounting a semiconductor integrated circuit (IC) chip. The die attach pad **220** is surrounded by the plurality of lead pads **210**. The die attach pad **220** is characterized by a rectangular shape. The die attach pad **220** includes a first protruding portion **221** on a first side **220-1** of the die attach pad **220** that is coupled to a first lead pad **210-1**. The die attach pad **220** also includes a second protruding portion **222** on a second side **220-2** of the die attach pad **220** opposite the first side. The second protruding portion **222** is coupled to a second lead pad **210-2**. Otherwise, the die attach pad **220** does not comprise direct connections to the peripheral frame **201**, without going through a lead pad **210**.

[0034] FIG. 2 also shows cutline **205**, where, after the package is completed as shown in FIGS. 7 and 8, the connection portions **203** are cut off from the lead pads **210** to separate the lead pads **210** and the die attach pad **220** from the peripheral frame **201**.

[0035] In some embodiments, leadframe **200** also includes multiple lead pads on the first side **2200-1** of the die attach pad **220**, with two or more of the lead pads disposed on either side of the first lead pad. For example, as shown in FIG. 2, lead pads **210-11** and **210-12** are on the left side of the first lead pad **210-1**. Further, lead pads **210-13** and **210-14** are on the right side of the first lead pad **210-1**. Similarly, multiple lead pads are on the second side of the die attach pad, with two or more of the lead pads disposed on either side of the second lead pad. As shown in FIG. 2, lead pads **210-21** and **210-22** are on the left side of the first lead pad **210-2**. Further, lead pads **210-23** and **210-24** are on the right side of the first lead pad **210-2**.

[0036] In some embodiments, the die attach pad also includes an additional protruding portion on either side of the first protruding portion, coupled to a respective lead pad on either side of the first lead pad. The die attach pad also includes an additional protruding portion on either side of the second protruding portion, coupled to a respective lead pad on either side of the second lead pad. Examples of these embodiments are shown in FIGS. 11 and 12, as described below.

[0037] In some embodiments, the die attach pad is characterized by a rectangular shape having a length longer than a width. In these embodiments, the die attach pad also includes a third protruding portion on the first side of the die attach pad, the third protruding portion coupled to a third lead pad. The die attach pad also includes a fourth protruding portion on the second side of the die attach pad opposite the first side, the fourth protruding portion coupled to a fourth lead pad. Examples of these embodiments are shown in FIGS. 13-15, as described below.

[0038] In the embodiment shown in FIG. 2, the protruding portions **221** and **222** are characterized by slanted edges, for example, **225** and **226**. In some embodiments, the protruding portions of the die attach pad are characterized by straight and parallel edges. In other embodiments, the protruding

portions of the die attach pad are characterized by curved edges and can have different curvatures.

[0039] FIG. 3 is a simplified schematic diagram illustrating a top view of a semiconductor package including a semiconductor chip on a leadframe according to some embodiments of the invention. As shown in FIG. 3, a semiconductor package **300** includes a leadframe **200** and semiconductor integrated circuit (IC) die **350**, also referred to as an IC chip. Leadframe **200** in FIG. 3 is similar to leadframe **200** described above in connection to FIG. 2, and the same reference numerals are used in the description. As shown in FIGS. 2 and 3, a leadframe **200** includes a peripheral frame and a plurality of lead pads **210**. Each of the plurality of lead pads **210** is physically connected to the peripheral frame **201** by a respective connecting portion **203**. Leadframe **200** also has a die attach pad **220** configured for mounting a semiconductor integrated circuit (IC) chip. The die attach pad **220** is surrounded by the plurality of lead pads **210**. The die attach pad **220** is characterized by a rectangular shape. The die attach pad **220** includes a first protruding portion **221** on a first side **220-1** of the die attach pad **220**, that is coupled to a first lead pad **210-1**. The die attach pad **220** also includes a second protruding portion **222** on a second side **220-2** of the die attach pad **220** opposite the first side. The second protruding portion **222** is coupled to a second lead pad **210-2**. Otherwise, the die attach pad **220** does not comprise direct connections to the peripheral frame **201**, without going through a lead pad **210**.

[0040] FIGS. 2 and 3 also shows cutline **205**, where, after the package is completed as shown in FIGS. 7 and 8, the connection portions **203** are cut off from the lead pads **210** to separate the lead pads **210** and the die attach pad **220** from the peripheral frame **201**.

[0041] As shown, a die attach pad is surrounded by a plurality of lead pads, wherein the die attach pad is characterized by a rectangular shape. The die attach pad includes a first protruding portion on a first side of the die attach pad, the first protruding portion coupled to a first lead pad. Multiple lead pads on the first side of the die attach pad are aligned with the first lead pad, with two or more of the lead pads disposed on both sides of the first lead pad. The die attach pad also includes a second protruding portion on a second side of the die attach pad opposite the first side, the second protruding portion coupled to a second lead pad. Multiple lead pads on the second side of the die attach pad are aligned with the second lead pad, with two or more of the lead pads disposed on both sides of the second lead pad.

[0042] As shown in FIG. 3, a semiconductor integrated circuit (IC) die **350** is attached to the die attach pad **220** of the leadframe **200**. In some embodiments, the die attach pad provides electrical ground for the first lead pad and the second lead pad. Semiconductor package **300** also includes bonding wires **330** coupling bonding pads for active signals on the IC die to lead pads on the first side of the die attach pad that are not coupled to the die attach pad and bonding wires **340** coupling bonding pads for active signals on the IC die to lead pads on the second side of the die attach pad that are not coupled to the die attach pad.

[0043] In some embodiments, the active signals include control and data signals. The grounding of the first lead pad and the second lead pad between active electrical signals can provide shielding and reduce cross talk and interference.

[0044] In some embodiments, electrical grounding for the semiconductor IC die can be provided either through con-

ductive die attach material or by bonding wires coupling the first lead pad and the second lead pad to ground bonding pads on the IC die.

[0045] As shown in FIGS. 2 and 3, the leadframe includes peripheral frame 201 connected to the plurality of lead pads 210. The die attach pad is coupled to the peripheral frame through the first lead pad 210-1 and the second lead pad 210-2. However, the die attach pad does not have a direct connection to the peripheral frame. In this embodiment, the first lead pad 210-1 and the second lead pad 210-2 provide sufficient mechanical connection between the die attach pad and the peripheral frame. An additional advantage of this design is that it simplifies that lead frame design and manufacturing.

[0046] FIG. 4 illustrates a top perspective view of a leadframe after singulation according to some embodiments of the invention. FIG. 4 shows a top perspective view of a leadframe 400, which is an example of leadframe 200 in FIG. 2 after the peripheral frame 201 is removed at the cutline 205.

[0047] FIG. 5 illustrates a bottom perspective view of a leadframe after singulation according to some embodiments of the invention. FIG. 5 shows a bottom perspective view of a leadframe 500, which is an example of leadframe 200 in FIG. 2 after the peripheral frame 201 is removed at the cutline 205.

[0048] FIG. 6 illustrates a bottom plan view of a leadframe after singulation according to some embodiments of the invention. FIG. 6 shows a bottom plan view of a leadframe 600, which is an example of leadframe 200 in FIG. 2 after the peripheral frame 201 is removed at the cutline 205.

[0049] FIG. 7 illustrates a top perspective view of a semiconductor package according to some embodiments of the invention. FIG. 7 shows a top perspective view of a semiconductor package 700 with a molding 710 covering the semiconductor IC chip and the leadframe after singulation. The side edges of lead pads 210 are visible from outside the package.

[0050] FIG. 8 illustrates a bottom perspective view of a semiconductor package according to some embodiments of the invention. FIG. 8 shows a bottom perspective view of a semiconductor package 800 with a molding 810 covering the semiconductor IC chip and the leadframe after singulation. The side and bottom edges of lead pads 210 and the bottom surface of the die attach pad 220 are visible from outside the package.

[0051] FIG. 9 illustrates a top perspective view of another leadframe after singulation according to some embodiments of the invention. FIG. 9 shows a top perspective view of a leadframe 900, which is similar to leadframe 400 in FIG. 4. However, the die attach pad extensions 921 and 922 in leadframe 900 are characterized by a curved shape. In contrast, the die attach pad extensions in leadframe 400 (and die attach pad extensions 221 and 222) are characterized by a slanted line shape.

[0052] FIG. 10 illustrates a top perspective view of another leadframe after singulation according to some embodiments of the invention. FIG. 10 shows a top perspective view of a leadframe 1000, which is similar to leadframe 900 in FIG. 9. However, the die attach pad extensions 1021 and 1022 in leadframe 1000 are also characterized by a curved shape with a smaller radius of curvature than die attach pad extensions 921 and 922 in leadframe 900. It is noted that, depending on the embodi-

ments, the die attach pad extensions can also have other shapes. For example, die attach pad extensions 1221 and 1222 of leadframe 1200 shown below in FIG. 12 are characterized by straight parallel edges.

[0053] In some embodiments, the die attach pad also includes an additional protruding portion on either side of the first protruding portion, coupled to a respective lead pad on either side of the first lead pad. The die attach pad also includes an additional protruding portion on either side of the second protruding portion, coupled to a respective lead pad on either side of the second lead pad. Examples of these embodiments are shown in FIGS. 12 and 13, as explained below.

[0054] FIG. 11 illustrates a top perspective view of another leadframe after singulation according to some embodiments of the invention. In FIG. 11, leadframe 11 includes a plurality of lead pads 1110, each of the plurality of lead pads physically connected to a peripheral frame by a respective connecting portion (not shown, after singulation). A die attach pad 1120 is configured for mounting a semiconductor integrated circuit (IC) chip, the die attach pad being surrounded by the plurality of lead pads 1110. The die attach pad 1120 includes a first protruding portion 1121 on a first side of the die attach pad, the first protruding portion coupled to a first lead pad 1110-1. The die attach pad 1120 also includes a second protruding portion 1122 on a second side of the die attach pad opposite the first side, the second protruding portion 1122 coupled to a second lead pad 1110-2. The die attach pad does not comprise direct connections to the peripheral frame.

[0055] In the embodiment of FIG. 11, the die attach pad 1120 also includes an additional protruding portion, 1121-2 and 1121-3, respectively, on either side of the first protruding portion 1121, coupled to a respective lead pad on either side of the first lead pad 1121. The die attach pad 1120 also includes an additional protruding portion, 1122-2 and 1122-3, respectively, on either side of the second protruding portion 1122, coupled to a respective lead pad on either side of the second lead pad 1122.

[0056] FIG. 12 illustrates a top perspective view of another leadframe after singulation according to some embodiments of the invention. FIG. 12 illustrates a leadframe 1200 that is similar to leadframe 1100 in FIG. 11. One difference is that the die attach pad extension in FIG. 11, e.g., 1121, is characterized by slanted edges, whereas the die attach pad extension in FIG. 12, e.g., 1221, is characterized by straight and parallel edges. In some other embodiments, the die attach pads have curved edges, and the curved edges can have different curvatures.

[0057] In some embodiments, the die attach pad 1120 is characterized by a rectangular shape having a length longer than a width and is configured to mount two or more semiconductor IC chips. Some examples are illustrated in FIGS. 11-17.

[0058] FIG. 13 is a simplified schematic diagram illustrating a top view of a leadframe according to some embodiments of the invention. As illustrated in FIG. 13, a leadframe 1300 includes a peripheral frame 1301, a plurality of lead pads 1310, and a die attach pad 1320. Each of the plurality of lead pads is physically connected to the peripheral frame by a respective connecting portion 1303. The die attach pad 1320 is configured for mounting a semiconductor integrated circuit (IC) chip, and the die attach pad is surrounded by the plurality of lead pads. The die attach pad is characterized by

a rectangular shape, with a length greater than a width. The die attach pad 1320 includes a first protruding portion 1321 on a first side of the die attach pad 1320, that is coupled to a first lead pad 1310-1. The die attach pad 1320 also includes a second protruding portion 1322 on a second side of the die attach pad 1320 opposite the first side. The second protruding portion 1322 is coupled to a second lead pad 1310-2. Otherwise, the die attach pad 220 does not comprise direct connections to the peripheral frame 1301, without going through a lead pad 1310.

[0059] FIG. 13 also shows cut line 1305, where, after the package is completed as shown in FIGS. 7 and 8, the connection portions 1303 are cut off from the lead pads 1310 to separate the lead pads 1310 and the die attach pad 1320 from the peripheral frame 1301.

[0060] Leadframe 1300 also includes a third protruding portion 1323 on the first side of the die attach pad, the third protruding portion 1323 coupled to a third lead pad 1310-3. Leadframe 1300 also includes a fourth protruding portion 1324 on the second side of the die attach pad opposite the first side, the fourth protruding portion 1324 coupled to a fourth lead pad 1310-4.

[0061] Leadframe 1300 also has other features similar to those in leadframe 200 in FIG. 2. For example, leadframe 1300 further includes multiple lead pads on the first side of the die attach pad, with two or more of the lead pads disposed on either side of the first lead pad; and multiple lead pads on the second side of the die attach pad, with two or more of the lead pads disposed on either side of the second lead pad.

[0062] FIG. 14 is a simplified schematic diagram illustrating a top view of a semiconductor package including two semiconductor chips on a leadframe according to some embodiments of the invention. As shown in FIG. 14, semiconductor package 1400 includes a leadframe similar to leadframe 1300 in FIG. 13, and corresponding features are labeled with the same reference numerals. Semiconductor package 1400 includes first and second semiconductor integrated circuit (IC) dies 1450 and 1460 attached to die attach pad 1320.

[0063] In some embodiments, the die attach pad provides electrical ground for the first, second, third, and fourth lead pads. Semiconductor package 1400 also includes bonding wires coupling the bonding pads for active signals on the IC die to lead pads on the first side of the die attach pad that are not coupled to the die attach pad and bonding wires coupling bonding pads for active signals on the IC die to lead pads on the second side of the die attach pad that are not coupled to the die attach pad.

[0064] In some embodiments, the active signals include control and data signals. The grounding of the first lead pad and the second lead pad between active electrical signals can provide shielding and reduce cross talk and interference.

[0065] In some embodiments, electrical grounding for the semiconductor IC die can be provided either through conductive die attach material or by bonding wires coupling the first lead pad and the second lead pad to ground bonding pads on the IC die.

[0066] As shown in FIGS. 13 and 14, the leadframe includes peripheral frame 1301 connected to the plurality of lead pads 1310. The die attach pad is coupled to the peripheral frame through the first to fourth lead pads 1310-1, 1310-2, 1310-3, and 1310-4, but does not have a direct connection to the peripheral frame. In this embodiment, the

first to fourth lead pads 1310-1, 1310-2, 1310-3, and 1310-4, provide sufficient mechanical connection between the die attach pad and the peripheral frame. An additional advantage of this design is that it simplifies that lead frame design and manufacturing.

[0067] FIG. 15 illustrates a top perspective view of a leadframe after singulation according to some embodiments of the invention. FIG. 15 shows a top perspective view of a leadframe 1500, which is an example of leadframe 1300 in FIG. 13 after the peripheral frame 1301 is removed at the cutline 1305.

[0068] FIG. 16 illustrates a bottom perspective view of a leadframe after singulation according to some embodiments of the invention. FIG. 16 shows a bottom perspective view of a leadframe 1600, which is an example of leadframe 1300 in FIG. 13 after the peripheral frame 1301 is removed at the cutline 1305.

[0069] FIG. 17 illustrates a bottom perspective view of a semiconductor package according to some embodiments of the invention. FIG. 17 shows a bottom perspective view of a semiconductor package 1700 with a molding 1710 covering the semiconductor IC chip and the leadframe after singulation. The side and bottom edges of lead pads 1310 and the bottom surface of the die attach pad are visible from outside the package.

[0070] FIG. 18 is a simplified flowchart illustrating a method for forming a semiconductor package according to some embodiments of the invention. The flowchart in FIG. 18 describes a method 1800 for forming a semiconductor package, which is summarized here and further described below.

[0071] 1810—providing a leadframe;

[0072] 1820—attaching a first semiconductor integrated circuit (IC) die to the die attach pad;

[0073] 1830—bonding IC die bonding pads for active signals to lead pads;

[0074] 1840—applying a molding that covers portions of the leadframe, the die attach material, and the die; and

[0075] 1850—singulating the dies.

[0076] At 1810, the method includes providing a leadframe. In some embodiments, the leadframes can be formed in an array or a strip. For example, a sheet of alloy, such as a copper alloy, is patterned using an etching process or a stamping process. It may include an additional etching process and may be followed by a plating process for the plurality of leadframes.

[0077] In some embodiments, the lead frame includes a peripheral frame, a plurality of lead pads, and a die attach pad. Each of the plurality of lead pads is physically connected to the peripheral frame by a respective connecting portion. The die attach pad is configured for mounting a semiconductor integrated circuit (IC) chip. The die attach pad is surrounded by the plurality of lead pads. The die attach pad does not comprise direct connections to the peripheral frame. The die attach pad includes a first protruding portion on a first side of the die attach pad, the first protruding portion coupled to a first lead pad, and a second protruding portion on a second side of the die attach pad opposite the first side, the second protruding portion coupled to a second lead pad. Examples of the leadframe are described above in connections with FIGS. 1-17.

[0078] At 1820, the method includes attaching a first semiconductor integrated circuit (IC) die to the die attach

pad. In some embodiments, the semiconductor die is attached to a die attach pad by depositing a die-attach epoxy on each of the die attach pads, placing the semiconductor die on the epoxy, and then curing the epoxy.

[0079] At **1830**, the method includes bonding the IC die bonding pads for active signals to the lead pads on the first side of the die attach pad that are not coupled to the die attach pad. The method further includes bonding IC die bonding pads for active signals to lead pads on the second side of the die attach pad that are not coupled to the die attach pad. The semiconductor die is attached to the lead pads via wire bonds or other bonding methods.

[0080] At **1840**, the method includes applying a molding that covers portions of the leadframe, the die attach material, and the die. For example, a molding compound is applied to cover the semiconductor die and other components with respect to each of the leadframes.

[0081] At **1850**, the method includes separating the plurality of lead pads from the peripheral frame by cutting off the connecting portions between the lead pads and the peripheral frame. In some embodiments, the plurality of encapsulated leadframes in the array or strip of leadframes are marked and singulated into respective packages. Each of the packages are then tested.

[0082] In some embodiments of method **1800**, the leadframe also includes multiple lead pads on the first side of the die attach pad, with two or more of the lead pads disposed on either side of the first lead pad; and multiple lead pads on the second side of the die attach pad, with two or more of the lead pads disposed on either side of the second lead pad.

[0083] In some embodiments, the die attach pad also includes an additional protruding portion on either side of the first protruding portion, coupled to a respective lead pad on either side of the first lead pad; and an additional protruding portion on either side of the second protruding portion, coupled to a respective lead pad on either side of the second lead pad.

[0084] In some embodiments, the die attach pad is characterized by a rectangular shape having a length longer than a width. The die attach pad further includes a third protruding portion on the first side of the die attach pad, the third protruding portion coupled to a third lead pad; and a fourth protruding portion on the second side of the die attach pad opposite the first side, the fourth protruding portion coupled to a fourth lead pad.

[0085] In some embodiments, the protruding portions of the die attach pad are characterized by straight edges.

[0086] In some embodiments, the protruding portions of the die attach pad are characterized by curved edges.

[0087] The features of several embodiments are described above to highlight some aspects of the present disclosure. It is understood that the examples and embodiments described herein are for illustrative purposes only and that various modifications or changes in light thereof will be suggested to persons skilled in the art and are to be included within the spirit and purview of this disclosure.

What is claimed is:

1. A method for forming a semiconductor package, the method comprising:

providing a leadframe, the leadframe comprising:

a peripheral frame;

a plurality of lead pads, each of the plurality of lead pads physically connected to the peripheral frame by a respective connecting portion; and

a die attach pad configured for mounting a semiconductor integrated circuit (IC) chip, the die attach pad being surrounded by the plurality of lead pads, wherein the die attach pad is characterized by a rectangular shape and comprises:

a first protruding portion on a first side of the die attach pad, the first protruding portion being coupled to a first lead pad; and

a second protruding portion on a second side of the die attach pad opposite the first side, the second protruding portion being coupled to a second lead pad, wherein the die attach pad does not comprise direct connections to the peripheral frame;

attaching a first semiconductor IC die to the die attach pad;

bonding IC die bonding pads for active signals to lead pads on the first side of the die attach pad that are not coupled to the die attach pad;

bonding IC die bonding pads for active signals to lead pads on the second side of the die attach pad that are not coupled to the die attach pad;

applying a molding that covers portions of the leadframe, a die attach material, and the first semiconductor IC die; and

separating the plurality of lead pads from the peripheral frame by cutting off connecting portions between the lead pads and the peripheral frame.

2. The method of claim 1, wherein the plurality of lead pads comprises:

a first plurality of lead pads on the first side of the die attach pad, with two or more of the first plurality of lead pads disposed on either side of the first lead pad; and

a second plurality of lead pads on the second side of the die attach pad, with two or more of the second plurality of lead pads disposed on either side of the second lead pad.

3. The method of claim 1, wherein the die attach pad further comprises:

a first additional protruding portion on either side of the first protruding portion, coupled to a first respective lead pad on either side of the first lead pad; and

a second additional protruding portion on either side of the second protruding portion, coupled to a second respective lead pad on either side of the second lead pad.

4. The method of claim 3, wherein the die attach pad is characterized by a rectangular shape having a length longer than a width, the die attach pad further including:

a third protruding portion on the first side of the die attach pad, the third protruding portion coupled to a third lead pad; and

a fourth protruding portion on the second side of the die attach pad opposite the first side, the fourth protruding portion coupled to a fourth lead pad.

5. The method of claim 1, wherein the first protruding portion and the second protruding portion of the die attach pad are characterized by straight edges.

6. The method of claim 1, wherein the first protruding portion and the second protruding portion of the die attach pad are characterized by curved edges.

7. The method of claim 1, wherein the die attach pad comprises a first portion of the die attach pad operable to accommodate the first semiconductor IC die attached to the

first portion of the die attach pad, and the die attach pad provides electrical ground for the first lead pad and the second lead pad.

8. The method of claim 7, further comprising:

coupling, using a first plurality of bonding wires, bonding pads for active signals on the semiconductor IC die to a first plurality of lead pads of the plurality of lead pads on the first side of the die attach pad that are not coupled to the die attach pad; and

coupling, using a second plurality of bonding wires, bonding pads for active signals on the semiconductor IC die to a second plurality of lead pads of the plurality of lead pads on the second side of the die attach pad that are not coupled to the die attach pad.

9. The method of claim 7, wherein the die attach pad further comprises a second portion of the die attach pad operable to accommodate a second semiconductor IC die attached to the second portion of the die attach pad.

10. The method of claim 9, wherein the first portion of the die attach pad and the second portion of the die attach pad are disposed laterally with respect to each other.

11. The method of claim 9, wherein a first area of the first portion of the die attach pad is substantially identical to a second area of the second portion of the die attach pad.

12. A method comprising:

providing a leadframe comprising a die attach pad surrounded by a plurality of lead pads, wherein the die attach pad is characterized by a rectangular shape and comprises:

a first protruding portion on a first side of the die attach pad, the first protruding portion coupled to a first lead pad;

a first plurality of lead pads on the first side of the die attach pad aligned with the first lead pad, with two or more of the first plurality of lead pads disposed on both sides of the first lead pad;

a second protruding portion on a second side of the die attach pad opposite the first side, the second protruding portion coupled to a second lead pad; and

a second plurality of lead pads on the second side of the die attach pad aligned with the second lead pad, with two or more of the second plurality of lead pads disposed on both sides of the second lead pad;

attaching a semiconductor integrated circuit (IC) die to the die attach pad of the leadframe, wherein the die attach pad provides electrical ground for the first lead pad and the second lead pad;

coupling, using a first plurality of bonding wires, bonding pads for active signals on the semiconductor IC die to the first plurality of lead pads on the first side of the die attach pad that are not coupled to the die attach pad; and coupling, using a second plurality of bonding wires, bonding pads for active signals on the semiconductor IC die to the second plurality of lead pads on the second side of the die attach pad that are not coupled to the die attach pad.

13. The method of claim 12, wherein the leadframe further comprises a peripheral frame connected to the first plurality of lead pads and the second plurality of lead pads,

wherein the die attach pad does not comprise a direct connection to the peripheral frame.

14. The method of claim 12, wherein the die attach pad further comprises:

a first additional protruding portion on either side of the first protruding portion, coupled to a first respective lead pad on either side of the first lead pad; and

a second additional protruding portion on either side of the second protruding portion, coupled to a second respective lead pad on either side of the second lead pad.

15. The method of claim 12, wherein the die attach pad is characterized by a rectangular shape having a length longer than a width, the die attach pad further comprising:

a third protruding portion on the first side of the die attach pad, the third protruding portion coupled to a third lead pad; and

a fourth protruding portion on the second side of the die attach pad opposite the first side, the fourth protruding portion coupled to a fourth lead pad.

16. The method of claim 15, further comprising:

attaching a second semiconductor IC die to the die attach pad of the leadframe;

coupling, using a third plurality of bonding wires, bonding pads for ground contact on the second semiconductor IC die to the third lead pad and the fourth lead pad, respectively;

coupling, using a fourth plurality of bonding wires, bonding pads for active signals on the second semiconductor IC die to the first plurality of lead pads on the first side of the die attach pad, with two or more of the first plurality of lead pads disposed on both sides of the third lead pad; and

coupling, using a fifth plurality of bonding wires, bonding pads for active signals on the second semiconductor IC die to the second plurality of lead pads on the second side of the die attach pad, with two or more of the second plurality of lead pads disposed on both sides of the fourth lead pad.

17. The method of claim 12, wherein the first protruding portion and the second protruding portion of the die attach pad are characterized by straight edges.

18. The method of claim 12, wherein the first protruding portion and the second protruding portion of the die attach pad are characterized by curved edges.

19. The method of claim 12, further comprising applying a molding that covers portions of the leadframe, a die attach material, and the semiconductor IC die, wherein a portion of the die attach pad, the first plurality of lead pads, and the second plurality of lead pads are exposed.

20. The method of claim 19, further comprising separating the first plurality of lead pads and the second plurality of lead pads from a peripheral frame by cutting off connecting portions connecting the first plurality of lead pads and the second plurality of lead pads to the peripheral frame.

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